



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	117	Media
HackerNews	6	Community
TechCrunch AI	3	Media
The Verge AI	1	Media

VERIFIED INTELLIGENCE — 1 story

VERI

When Large Language Models Fail in Healthcare: Evaluating Sensitivity to Prompt Variations

ArXiv CS.AI +1 source

80%

arXiv:2606.07237v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly used in healthcare for tasks such as clinical question answering, diagnosis support, and report summarization. Despite their promise, these models remain highly sensitive to subtle...

<https://arxiv.org/abs/2606.07237>

CONFIRMED SIGNALS — 125 stories

CONF

Show HN: Lathe - Use LLMs to learn a new domain, not skip past it

HackerNews 60% HN 303

Hey HN! Lathe is an experiment in using LLMs to teach me something new, instead of doing the work for me. It generates a hands-on, source-backed tutorial for any technical topic you want to learn. Then you work through it yourself by reading and typing the code by hand (gasp)...

<https://github.com/devenjarvis/lathe>

CONF

Show HN: Nightwatch, The open-source, read-only AI SRE

HackerNews 50% HN 17

nightwatch is a local-first, read-only layer on top of your monitoring. it groups alert storm into incidents, flags noisy checks and has an agent that can investigate for you live systems. You can e.g. jump from the incident into the agent directly. the reason for this weekend...

<https://github.com/ninoxAI/nightwatch>

CONF

Show HN: Create and Maintain Filesystem Structures for LLMs [v1.0.11 Out]

HackerNews 50% HN 5

No summary — see link for full article.

<https://pypi.org/project/seed-cli/>

CONF

Show HN: Fact check, summarize, explain any article or webpage on local CPU

HackerNews 50% HN 5

No summary — see link for full article.

<https://github.com/kouhxp/fftext>

CONF

Show HN: Web Speed - A shared web-map registry for AI agents (MCP, open source)

HackerNews 50% HN 4

Hey everyone, I've been working on a tool called Web Speed. It takes web pages full of HTML and parses them into an easily readable sitemap for an AI agent. By ensuring the AI doesn't have to analyze the full HTML from the page or screenshots of the page, it makes the agent a...

<https://www.getwebspeed.io/>

CONF

Show HN: Persist - an AI agent that follows up accrued channels till they reply

HackerNews 50% HN 4

Hi HN, I am co-founder of Persist AI. Meet Persist.chat an AI sales agent that follows up to prospects till you secure a sale. It can help you find prospects for your product with our extensive lists of LinkedIn and email lists and also you can bring your own list of customers...

<https://persist.chat>

CONF

Notion restores access to Anthropic after service disruption

TechCrunch AI 50%

Notion's head of product said he was "astonished" at "the amount of people RT-ing this."

<https://techcrunch.com/2026/06/07/notion-restores-access-to-anthropic-after-service-dis...>

CONF

OpenAI is still working on that 'super app'

TechCrunch AI 50%

"Chat is dead" — at least, according to a senior OpenAI employee.

<https://techcrunch.com/2026/06/07/openai-is-still-working-on-that-super-app/>

0 CONF

AI 'content creators' are getting harder to spot

The Verge AI 50%

This is The Stepback, a weekly newsletter breaking down one essential story from the tech world. For more on AI confusion, follow Robert Hart. The Stepback arrives in our subscribers' inboxes at 8AM ET. Opt in for The Stepback here. How it started At first, AI influencers were...

<https://www.theverge.com/ai-artificial-intelligence/943187/ai-content-creators>

1 CONF

DiBS: Diffusion-Informed Branch Selection

ArXiv CS.AI 50%

arXiv:2606.06518v1 Announce Type: new Abstract: Sudoku is a representative constraint satisfaction problem that requires global structural reasoning under strict discrete constraints. The existing works of solving Sudoku mainly focus on two dominant approaches, i.e., traditional...

<https://arxiv.org/abs/2606.06518>

2 CONF

SafeGene: Reusable Adapters for Transferable Safety Alignment

ArXiv CS.AI 50%

arXiv:2606.06519v1 Announce Type: new Abstract: Open-weight LLMs are increasingly fine-tuned into customized assistants, but downstream fine-tuning can weaken safety alignment and make models more vulnerable to malicious prompts, even when the training data is not intentionally...

<https://arxiv.org/abs/2606.06519>

3 CONF

Attack Selection in Agentic AI Control Evaluations Meaningfully Decreases Safety

ArXiv CS.AI 50%

arXiv:2606.06529v1 Announce Type: new Abstract: An attacker that strategically chooses when to attack is much harder to catch than one that attacks indiscriminately. AI control is a safety framework for deploying capable but untrusted AI agents under the oversight of a weaker,...

<https://arxiv.org/abs/2606.06529>

4 **CONF****CARVE-Q: Quantum-Proposed, Classically Certified Interactive Driving Repair****ArXiv CS.AI 50%**

arXiv:2606.06531v1 Announce Type: new Abstract: The critical question after a correct driving veto is not only whether a maneuver is unsafe, but whether the blocked interaction admits a lawful, auditable, and responsibility-bounded repair. Prediction and game-theoretic planners...

<https://arxiv.org/abs/2606.06531>

5 **CONF****Position: Don't Just "Fix it in Post": A Science of AI Must Study Training Dynamics****ArXiv CS.AI 50%**

arXiv:2606.06533v1 Announce Type: new Abstract: What would it mean to have a scientific understanding of AI? Models are not static objects: they are snapshots of time-evolving processes shaped by data, objectives, architectures, and optimization dynamics. Yet much of AI research...

<https://arxiv.org/abs/2606.06533>

6 **CONF****AEGIS: A Backup Reflex for Physical AI****ArXiv CS.AI 50%**

arXiv:2606.06660v1 Announce Type: new Abstract: Long-horizon robot manipulation tends to fail gradually: one bad step degrades the state, and the policy spirals into a basin from which it cannot recover. The failure is often visible before it happens. We introduce AEGIS...

<https://arxiv.org/abs/2606.06660>

7 **CONF****OpenSkill: Open-World Self-Evolution for LLM Agents****ArXiv CS.AI 50%**

arXiv:2606.06741v1 Announce Type: new Abstract: Self-evolving agents requires adaptation after deployment, but existing approaches assume a usable learning loop, such as curated skills, successful trajectories, or verifier signals. Real open-world deployments may provide none of...

<https://arxiv.org/abs/2606.06741>

8 **CONF****Evidence-Based Intelligent Diagnostic and Therapeutic Visualization System with Large Language Models: Multi-Turn Interaction and Multimodal Treatment Plan Generation****ArXiv CS.AI 50%**

arXiv:2606.06869v1 Announce Type: new Abstract: Aim: Existing AI-assisted traditional Chinese medicine diagnostic tools suffer from opaque reasoning processes, passive interaction, and limited treatment plan presentation. This study proposes a knowledge-enhanced visual...

<https://arxiv.org/abs/2606.06869>

9 **CONF****Declarative Skills for AI Agents in Knowledge-Grounded Tool-Use Workflows****ArXiv CS.AI 50%**

arXiv:2606.06923v1 Announce Type: new Abstract: We study orchestration mechanisms for tool-using AI agents in realistic customer-service workflows over an unstructured knowledge base. We argue that declarative agents -- AI agents equipped with natural-language skill files...

<https://arxiv.org/abs/2606.06923>

0 CONF

Accounting for Context: Shaping Moral Credences for Value Alignment

ArXiv CS.AI 50%

arXiv:2606.06972v1 Announce Type: new Abstract: Ensuring that agent behaviours are aligned with human moral values inevitably raises the problem of how to account for the plurality of moral perspectives that societies -- and even individuals -- typically adopt. Work on moral...

<https://arxiv.org/abs/2606.06972>

1 CONF

Exploring Agentic Tool-Calling Decisions via Uncertainty-Aligned Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2606.06976v1 Announce Type: new Abstract: Large language model (LLM)-based agents often make suboptimal tool-use decisions, including unsupported tool invocation and hallucinated direct responses, which may accumulate errors throughout multi-step interactions. Existing...

<https://arxiv.org/abs/2606.06976>

2 CONF

Teaching the Way, Not the Answer: Privileged Tutoring Distillation for Multimodal Policy Optimization

ArXiv CS.AI 50%

arXiv:2606.07000v1 Announce Type: new Abstract: Recent post-training methods, particularly Reinforcement Learning with Verifiable Rewards (RLVR), have significantly enhanced the reasoning ability of Large Vision-Language Models (LVLMs). However, the sparse nature of verifiable...

<https://arxiv.org/abs/2606.07000>

3 CONF

The Sim-to-Real Gap of Foundation Model Agents: A Unified MDP Perspective

ArXiv CS.AI 50%

arXiv:2606.07017v1 Announce Type: new Abstract: Foundation model agents are increasingly deployed for real-world decision-making, but suffer from the sim-to-real gap. While robotics and classical control have mature frameworks to address this gap, the foundation model community...

<https://arxiv.org/abs/2606.07017>

4 CONF

StainFlow: Entity-Stain Tracking and Evidence Linking for Process Rewards in GUI Agents

ArXiv CS.AI 50%

arXiv:2606.07027v1 Announce Type: new Abstract: Reinforcement Learning (RL) has become a promising approach for improving GUI Agents in long-horizon, stochastic digital environments, but trajectory-level success feedback is too sparse to provide reliable credit assignment for...

<https://arxiv.org/abs/2606.07027>

5 CONF

Hierarchical Certified Semantic Commitment for Byzantine-Resilient LLM-Agent Collaboration

ArXiv CS.AI 50%

arXiv:2606.07316v1 Announce Type: cross Abstract: Byzantine collaboration among large-language-model agents requires a finality-control primitive: given delivered stochastic, structured natural-language proposals, the protocol must decide whether the round supports a commit,...

<https://arxiv.org/abs/2606.07316>

6 CONF

Think Fast: Estimating No-CoT Task-Completion Time Horizons of Frontier AI Models

ArXiv CS.AI 50%

arXiv:2606.07157v1 Announce Type: new Abstract: Many efforts to ensure frontier AI models are safe rely on monitoring their chain-of-thought (CoT) reasoning. If models become able to perform sufficiently complex reasoning internally, without explicit thinking tokens, this would...

<https://arxiv.org/abs/2606.07157>

7 CONF

Online Pandora's Box for Contextual LLM Cascading

ArXiv CS.AI 50%

arXiv:2606.07392v1 Announce Type: new Abstract: Motivated by Large Language Model (LLM) cascading, we propose an online contextual Pandora's Box model for adaptively querying and selecting LLM APIs. In each period, a decision-maker observes a request context and faces a...

<https://arxiv.org/abs/2606.07392>

8 CONF

MacArena: Benchmarking Computer Use Agents on an Online macOS Environment

ArXiv CS.AI 50%

arXiv:2606.06560v1 Announce Type: cross Abstract: Computer-use agents (CUAs) operate graphical user interfaces (GUIs) through vision and control primitives, and their capabilities have advanced rapidly, driven in part by standardized online evaluation benchmarks such as OSWorld,...

<https://arxiv.org/abs/2606.06560>

9 CONF

How AI Agents Reshape Knowledge Work: Autonomy, Efficiency, and Scope

ArXiv CS.AI 50%

arXiv:2606.07489v1 Announce Type: new Abstract: Frontier AI systems are bridging the gap between intelligence and utility by shifting from conversational assistants to autonomous agents that execute tasks end to end. Using production data from Perplexity's Search and Computer...

<https://arxiv.org/abs/2606.07489>

0 CONF

Zero-Shot Embedding Drift Detection: A Lightweight Defense Against Prompt Injections in LLMs

ArXiv CS.AI 50%

arXiv:2601.12359v1 Announce Type: cross Abstract: Prompt injection attacks have become an increasing vulnerability for LLM applications, where adversarial prompts exploit indirect input channels such as emails or user-generated content to circumvent alignment safeguards and...

<https://arxiv.org/abs/2601.12359>

1 CONF

Trading Engagement for Sustainability: Carbon-Aware Re-ranking for E-commerce Recommendations

ArXiv CS.AI 50%

arXiv:2606.04550v1 Announce Type: cross Abstract: E-commerce recommender systems strongly influence which products users consider and purchase, yet sustainability signals such as Product Carbon Footprint (PCF) are almost never available at catalog scale. We study carbon-aware...

<https://arxiv.org/abs/2606.04550>

2 CONF

Human Adults and LLMs as Scientists: Who Benefits from Active Exploration?

ArXiv CS.AI 50%

arXiv:2606.06464v1 Announce Type: cross Abstract: A long-standing finding in the causal learning literature is that adults struggle to identify conjunctive causal rules, where an effect requires the simultaneous presence of multiple causes, while performing better in disjunctive...

<https://arxiv.org/abs/2606.06464>

3 CONF

Which Anatomy Matters Under Limited Labels? A Data-Efficient Anatomy-Aware Benchmark for Cardiac Pathology Prediction

ArXiv CS.AI 50%

arXiv:2606.06509v1 Announce Type: cross Abstract: Numerous medical imaging problems must be solved under limited labels and constrained compute, yet it remains unclear whether performance gains are driven mainly by more expressive models or by better representation of clinically...

<https://arxiv.org/abs/2606.06509>

4 CONF

FP8 is All You Need (Part 1): Debunking Hardware FP64 as the HPC Holy Grail

ArXiv CS.AI 50%

arXiv:2606.06510v1 Announce Type: cross Abstract: Conventional HPC dogma holds that native hardware FP64 silicon is the irreducible foundation of scientific computing -- the "holy grail" of double-precision simulation. This paper argues the dogma is wrong: on AI-optimised GPUs...

<https://arxiv.org/abs/2606.06510>

5 CONF

DxPTA: An Architecture Design Space Exploration with Optical Dataflow-guided Strategy for HW/SW Co-Design of Photonic Transformer Accelerators

ArXiv CS.AI 50%

arXiv:2606.06515v1 Announce Type: cross Abstract: Transformer-based networks have emerged as prominent AI models with state-of-the-art performance, which potentially pave the way toward artificial general intelligence (AGI). However, their large sizes still hinder their...

<https://arxiv.org/abs/2606.06515>

6 CONF

Benchmarking Language Modeling for Lossless Compression of Full-Fidelity Audio

ArXiv CS.AI 50%

arXiv:2603.08683v2 Announce Type: replace-cross Abstract: Autoregressive "language" models (LMs) trained on raw waveforms can be repurposed for lossless audio compression, but prior work is limited to 8-bit audio, leaving open whether such approaches work for practical settings...

<https://arxiv.org/abs/2603.08683>

7 CONF

Attention Consistent Longitudinal Medical Visual Question Answering Guided by Vision Foundation Models

ArXiv CS.AI 50%

arXiv:2606.06534v1 Announce Type: cross Abstract: Longitudinal medical visual question answering (VQA) requires reasoning about anatomical differences between an image of a current time point and an image of a referred time point. We propose an attention-guided encoder-decoder...

<https://arxiv.org/abs/2606.06534>

8 **CONF****Synthetic Benchmarks Overstate Forward-Forward Scaling: Real-Data Limits of Layer-Local Training**

ArXiv CS.AI 50%

arXiv:2606.06539v1 Announce Type: cross Abstract: Forward-Forward (FF) learning [Hinton, 2022] replaces backpropagation with strictly layer-local goodness updates. Recent FF-CNN work has narrowed the gap to BP on 32x32 benchmarks, raising the question of whether layer-local...

<https://arxiv.org/abs/2606.06539>

9 **CONF****Coordinated optimization of departure sequencing and section-track allocation in railway short-term concentrated departure scenarios based on qubo and hybrid quantum algorithms**

ArXiv CS.AI 50%

arXiv:2606.06543v1 Announce Type: cross Abstract: This study examines the coordinated optimization of departure sequencing and section-track allocation in railway short-term concentrated departure scenarios. A quadratic unconstrained binary optimization (QUBO) model is...

<https://arxiv.org/abs/2606.06543>

0 **CONF****FAIR-Calib: Frontier-Aware Instability-Rewighted Calibration for Post-Training Quantization of Diffusion Large Language Models**

ArXiv CS.AI 50%

arXiv:2606.06547v1 Announce Type: cross Abstract: Diffusion Large Language Models (dLLMs) refine tokens iteratively but commit them irreversibly, leading to a "stability lag" where early decisions remain fragile even after being written. We reveal that Post-Training Quantization...

<https://arxiv.org/abs/2606.06547>

1 **CONF****AI-Driven Test Case Generation from Natural Language Requirements: A Survey of Techniques and Research Gaps**

ArXiv CS.AI 50%

arXiv:2606.06563v1 Announce Type: cross Abstract: Software testing is critical for verifying that systems meet specified requirements, yet remains among the most time-consuming and expensive activities in development. Requirements-based test generation allows test cases to be...

<https://arxiv.org/abs/2606.06563>

2 **CONF****WAV: Multi-Resolution Block Residual Routing for Deep Decoder-Only Transformers**

ArXiv CS.AI 50%

arXiv:2606.06564v1 Announce Type: cross Abstract: Residual connections are central to training deep Transformers, but standard PreNorm residual streams aggregate sublayer updates with fixed unit weights. Recent Attention Residuals replace this fixed accumulation with...

<https://arxiv.org/abs/2606.06564>

3 **CONF****MalTree: Tracing Malware Evolution from Embeddings at Scale**

ArXiv CS.AI 50%

arXiv:2606.06570v1 Announce Type: cross Abstract: Malware detection remains largely reactive: machine learning models trained on known samples degrade as threats evolve. Understanding evolutionary relationships among malware families can inform proactive defense, but traditional...

<https://arxiv.org/abs/2606.06570>

4 **CONF****Generative Models Erode Human Temporal Learning Through Market Selection****ArXiv CS.AI** **50%**

arXiv:2606.06572v1 Announce Type: cross Abstract: We argue that modern generative models create structural risks for knowledge and cultural production at current, sub-AGI capability levels. We define Human Temporal Learning (HTL) as path-dependent knowledge accumulation through...

<https://arxiv.org/abs/2606.06572>

5 **CONF****Re-Centering Humans in LLM Personalization****ArXiv CS.AI** **50%**

arXiv:2606.06614v1 Announce Type: cross Abstract: Despite growing interest, most evaluations of large language models' (LLMs') personalization abilities have relied on synthetic data. It remains unclear how well current personalization systems work for real users. In this paper,...

<https://arxiv.org/abs/2606.06614>

6 **CONF****FIGMA: Towards Fine-Grained Music retrieval****ArXiv CS.AI** **50%**

arXiv:2606.06615v1 Announce Type: cross Abstract: Retrieving music using natural language descriptions has improved with contrastive audio-text models such as CLAP, but current systems remain limited to coarse semantic queries. When descriptions specify fine-grained musical...

<https://arxiv.org/abs/2606.06615>

7 **CONF****ChronoForest: Closed-Loop Multi-Tree Diffusion Planning for Efficient Bridge Search and Route Composition****ArXiv CS.AI** **50%**

arXiv:2606.06618v1 Announce Type: cross Abstract: How can we plan long-horizon routes that reach designated goals, visit required waypoints, and remain short when only short-horizon offline trajectories are available? This problem matters in offline navigation because collecting...

<https://arxiv.org/abs/2606.06618>

8 **CONF****When is 3D Worth It? A Resource-Performance Frontier for CNNs and Transformers in Lung CT****ArXiv CS.AI** **50%**

arXiv:2606.06950v1 Announce Type: cross Abstract: Three-dimensional models are widely assumed preferable for volumetric medical imaging, yet their practical value depends on whether performance gains justify added computational cost and complexity. Rather than proposing a new...

<https://arxiv.org/abs/2606.06950>

9 **CONF****The Geography of Algorithmic Judgment: LLM Intermediaries, Place Identity, and Racial Steering in Housing Search****ArXiv CS.AI** **50%**

arXiv:2606.06694v1 Announce Type: cross Abstract: Large language models (LLMs) are rapidly assuming an intermediary role in housing search through the integration of listing platforms within conversational interfaces, mediating access to information, search, and recommendations...

<https://arxiv.org/abs/2606.06694>

- 0
CONF
SpectCount: Spectrotemporal Counting via Synthetic Signals Improves Large Audio Language Models
ArXiv CS.AI 50%
arXiv:2606.06907v1 Announce Type: cross Abstract: Large audio language models (LALMs) extend large language models with an audio encoder and large-scale audio data. However, the scarcity of high-quality annotated audio data remains a fundamental bottleneck for scaling. Through...
<https://arxiv.org/abs/2606.06907>
- 1
CONF
Acoustic Cue Alignment in Audio Language Models for Speech Emotion Recognition
ArXiv CS.AI 50%
arXiv:2606.07309v1 Announce Type: cross Abstract: Instruction-following audio language models (ALMs) can be augmented with explicit acoustic cues, yet it remains unclear whether such cues are used in a grounded way when the raw audio is already available. We study this question...
<https://arxiv.org/abs/2606.07309>
- 2
CONF
Does Topic Sentiment Cause Perceived Ideology? Comparing Human and LLM Annotations in Political News Articles
ArXiv CS.AI 50%
arXiv:2606.06715v1 Announce Type: cross Abstract: We ask whether topic sentiment has a causal effect on perceived political ideology, and whether the answer depends on who assigns the ideology label. Using articles from AllSides, paired with shared sentiment annotations from...
<https://arxiv.org/abs/2606.06715>
- 3
CONF
ViVa: A Video-Generative Value Model for Robot Reinforcement Learning
ArXiv CS.AI 50%
arXiv:2604.08168v2 Announce Type: replace-cross Abstract: Vision-language-action (VLA) models have advanced robot manipulation through large-scale pretraining, but real-world deployment remains challenging due to partial observability and delayed feedback. Reinforcement learning...
<https://arxiv.org/abs/2604.08168>
- 4
CONF
MSAIC-Net: A Multi-Scale Attention and Imbalance-Aware Contrastive Network for ECG-Based Myocardial Substrate Abnormality Detection
ArXiv CS.AI 50%
arXiv:2606.06718v1 Announce Type: cross Abstract: Myocardial substrate abnormalities, such as myocardial scar and myocardial infarction (MI), are associated with adverse cardiovascular outcomes. Electrocardiography (ECG) provides a low-cost and widely available tool for...
<https://arxiv.org/abs/2606.06718>
- 5
CONF
SCOUT: Semantic scene COverage via Uncertainty-guided Traversal
ArXiv CS.AI 50%
arXiv:2606.06721v1 Announce Type: cross Abstract: Robots that operate over extended periods should not merely visit space; they should progressively understand it. Yet most 3D scene graph pipelines treat perception as a post-processing stage over a fixed dataset, decoupling...
<https://arxiv.org/abs/2606.06721>

6 CONF

SV-Detect: AI-generated Text Detection with Steering Vectors

ArXiv CS.AI 50%

arXiv:2606.07313v1 Announce Type: cross Abstract: Detecting machine-generated text is especially difficult under distribution shift, such as transfer across domains, source models, and editing attacks. We propose a fake-text detector based on steering vectors extracted from the...

<https://arxiv.org/abs/2606.07313>

7 CONF

What Your Posts Reveal: A Benchmark and Agentic Framework for User-Level Privacy Leakage on Social Media

ArXiv CS.AI 50%

arXiv:2606.06784v1 Announce Type: cross Abstract: Public social media posts can reveal private information through weak cues scattered across text, images, or metadata. Such leakage is often cumulative and cross-post: cues that appear harmless in isolation may jointly expose a...

<https://arxiv.org/abs/2606.06784>

8 CONF

Reinforcement Learning from Rich Feedback with Distributional DAgger

ArXiv CS.AI 50%

arXiv:2606.05152v2 Announce Type: replace-cross Abstract: Reasoning models have advanced rapidly, but the dominant reinforcement learning from verifiable rewards (RLVR) recipe remains surprisingly narrow: sample many responses and reward each with a single bit indicating whether...

<https://arxiv.org/abs/2606.05152>

9 CONF

PandaAI: A Practical Agent CQ2 for Neuro-symbolic Data Analysis And Integrated Decision-Making in Quantitative Finance

ArXiv CS.AI 50%

arXiv:2606.06823v1 Announce Type: cross Abstract: While deep learning has excelled in various domains, its application to sequential decision-making in finance remains challenging due to the low Signal-to-Noise Ratio (SNR) and non-stationarity of financial data. Leveraging the...

<https://arxiv.org/abs/2606.06823>

10 CONF

Progress-SQL: Improving Reinforcement Learning for Text-to-SQL via Progressive Rewards

ArXiv CS.AI 50%

arXiv:2606.06825v1 Announce Type: cross Abstract: Reinforcement learning has recently shown promise in improving large language models for Text-to-SQL generation, yet existing methods typically optimize one-shot rewards defined over a single SQL state. Such rewards provide...

<https://arxiv.org/abs/2606.06825>

11 CONF

Think Like a Pilot: Fine-Grained Long-Horizon UAV Navigation

ArXiv CS.AI 50%

arXiv:2606.06836v1 Announce Type: cross Abstract: Language-guided UAV agents must execute long-horizon semantic instructions while producing smooth, physically feasible continuous flight commands, yet existing Vision-Language Navigation (VLN) benchmarks typically use discrete or...

<https://arxiv.org/abs/2606.06836>

2 CONF

LLM Agent-Assisted Reverse Engineering with Quantitative Readability Metrics

ArXiv CS.AI 50%

arXiv:2606.06838v1 Announce Type: cross Abstract: Automatic decompilers produce functionally correct but often unreadable C code. This paper addresses one stage of the reverse engineering workflow: improving the readability of decompiled code using LLM agents guided by...

<https://arxiv.org/abs/2606.06838>

3 CONF

Textual Supervision Enhances Geospatial Representations in Vision-Language Models

ArXiv CS.AI 50%

arXiv:2606.07172v1 Announce Type: cross Abstract: Geospatial understanding is a critical yet underexplored dimension in the development of machine learning systems for tasks such as image geolocation and spatial reasoning. In this work, we analyze the geospatial representations...

<https://arxiv.org/abs/2606.07172>

4 CONF

Neuro-Symbolic Learning for Long-Horizon Task Planning Under Complex Logical Constraints

ArXiv CS.AI 50%

arXiv:2606.06877v1 Announce Type: cross Abstract: Task planning often suffers from severe efficiency bottlenecks when robots must reason over long-horizon action sequences under complex logical constraints, including object affordances, spatial relationships, and sequential...

<https://arxiv.org/abs/2606.06877>

5 CONF

FreeAnimate: Training-Free Human Image Animation with Preview-Guided Denoising

ArXiv CS.AI 50%

arXiv:2606.06885v1 Announce Type: cross Abstract: Human Image Animation has seen significant advancements, primarily driven by diffusion models. However, existing methods typically demand substantial training data and resources to achieve high-quality results, limiting...

<https://arxiv.org/abs/2606.06885>

6 CONF

Beyond Skeletons: Learning Animation Directly from Driving Videos with Same2X Training Strategy

ArXiv CS.AI 50%

arXiv:2606.06903v1 Announce Type: cross Abstract: Human image animation aims to generate a video from a static reference image, guided by pose information extracted from a driving video. Existing approaches often rely on pose estimators to extract intermediate representations,...

<https://arxiv.org/abs/2606.06903>

7 CONF

EASE-TTT: Evidence-Aligned Selective Test-Time Training for Long-Context Question Answering

ArXiv CS.AI 50%

arXiv:2606.06906v1 Announce Type: cross Abstract: Long-context question answering (QA) remains challenging for smaller language models even when answer-bearing evidence is already present in the input. Existing within-context retrieval methods localize and expose candidate...

<https://arxiv.org/abs/2606.06906>

8 CONF

A Comprehensive Anatomy of Human and DeepSeek-R1 LLM Mathematical Reasoning

ArXiv CS.AI 50%

arXiv:2606.07410v1 Announce Type: cross Abstract: The emergence of "Aha moments" in large language models, particularly DeepSeek-R1-0120, has raised the question of whether these systems genuinely reason or merely imitate the appearance of reasoning. We conduct a comprehensive...

<https://arxiv.org/abs/2606.07410>

9 CONF

Didact: A Cross-Domain Capability Discovery System for Defence

ArXiv CS.AI 50%

arXiv:2606.06942v1 Announce Type: cross Abstract: Policymakers in defence and defence-aligned sectors must monitor rapidly evolving research alongside sector priorities relevant to operational and strategic needs. In practice, these sources are fragmented across heterogeneous...

<https://arxiv.org/abs/2606.06942>

0 CONF

Auditing Training Data in Domain-adapted LLMs: LoRA-MINT

ArXiv CS.AI 50%

arXiv:2606.06946v1 Announce Type: cross Abstract: We present LoRA-MINT, a new methodology for Membership Inference Test (MINT) applied to recent Large Language Models (LLMs) fine-tuned for specific Natural Language Processing (NLP) tasks through Low-Rank Adaptation (LoRA). The...

<https://arxiv.org/abs/2606.06946>

1 CONF

OpenHalDet: A Unified Benchmark for Hallucination Detection across Diverse Generation Scenarios

ArXiv CS.AI 50%

arXiv:2606.06959v1 Announce Type: cross Abstract: Hallucination detection is essential for the reliable deployment of large language models (LLMs). However, existing evaluations face two core challenges: inconsistent inference configuration and evaluation, and limited coverage...

<https://arxiv.org/abs/2606.06959>

2 CONF

Never Seen Before: Benchmarking Genuine Zero-Shot Composed Image Retrieval with Consistent Video-Sourced Datasets

ArXiv CS.AI 50%

arXiv:2606.07032v1 Announce Type: cross Abstract: Zero-Shot Composed Image Retrieval (ZS-CIR) aims to retrieve a target image based on a query composed of a reference image and a relative caption without training samples. Existing ZS-CIR datasets often suffer from complete...

<https://arxiv.org/abs/2606.07032>

3 CONF

TRACE: Trajectory Reasoning through Adaptive Cross-Step Evidence Aggregation for LLM Agents

ArXiv CS.AI 50%

arXiv:2606.07054v1 Announce Type: cross Abstract: Autonomous LLM agents can pursue hidden malicious objectives through sequences of individually benign actions, making sabotage difficult to detect using standard trajectory-level monitoring. Existing approaches either evaluate...

<https://arxiv.org/abs/2606.07054>

4 **CONF****SlimSearcher: Training Efficiency-Aware Web Agents via Adaptive Reward Gating****ArXiv CS.AI 50%**

arXiv:2606.07074v1 Announce Type: cross Abstract: Deep research agents have demonstrated remarkable capabilities in complex information-seeking tasks, yet this power comes at a steep computational cost. Driven by accuracy-focused training paradigms, current models adopt...

<https://arxiv.org/abs/2606.07074>

5 **CONF****MetaConfigurator: AI-Assisted RDF Authoring from JSON Data****ArXiv CS.AI 50%**

arXiv:2606.07094v1 Announce Type: cross Abstract: Scientific workflows increasingly generate structured JSON data that is easy to exchange but difficult to interpret consistently across systems due to lacking semantic interoperability. While JSON Schema ensures structural...

<https://arxiv.org/abs/2606.07094>

6 **CONF****OffQ: Taming Structured Outliers in LLM Quantization by Offsetting****ArXiv CS.AI 50%**

arXiv:2606.07116v1 Announce Type: cross Abstract: Low-bit quantization has been widely adopted to accelerate the inference of large language models (LLMs) by significantly reducing computational cost and memory usage. However, activation outliers pose a major challenge to...

<https://arxiv.org/abs/2606.07116>

7 **CONF****Native3D: End-to-End 3D Scene Generation via Unified Mesh-Texture Modeling and Semantic Alignment****ArXiv CS.AI 50%**

arXiv:2606.07117v1 Announce Type: cross Abstract: This paper presents Native3D, the first end-to-end 3D scene generation framework that completely bypasses 2D intermediate representations. Traditional approaches typically require adapting 3D representations to the 2D domain to...

<https://arxiv.org/abs/2606.07117>

8 **CONF****REMEDI: A Benchmark for Retention and Unlearning Evaluation in Multi-label Clinical Disease Inference****ArXiv CS.AI 50%**

arXiv:2606.07141v1 Announce Type: cross Abstract: Language models trained for clinical disease inference are trained on patient data, which may include sensitive and private information, and data owners may request the removal of their data from a trained model due to privacy or...

<https://arxiv.org/abs/2606.07141>

9 **CONF****From Privacy to Workflow Integrity: Communication-Graph Metadata in Autonomous Agent Interoperability****ArXiv CS.AI 50%**

arXiv:2606.07150v1 Announce Type: cross Abstract: Agent-interoperability protocols such as A2A and MCP standardize what agents say to one another, but assume address-based transport over HTTP(S). Such transports protect message content, increasingly with end-to-end encryption....

<https://arxiv.org/abs/2606.07150>

0 CONF

UrduMMLU: A Massive Multitask Benchmark for Urdu Language Understanding

ArXiv CS.AI 50%

arXiv:2606.07167v1 Announce Type: cross Abstract: Meaningful multilingual evaluation must test models in the target language and educational context. Urdu, spoken by more than 230 million people, lacks a broad MMLU-style benchmark built from native educational sources. We...

<https://arxiv.org/abs/2606.07167>

1 CONF

An Abstract Architecture for Explainable Autonomy in Hazardous Environments

ArXiv CS.AI 50%

arXiv:2606.07211v1 Announce Type: cross Abstract: Autonomous robotic systems are being proposed for use in hazardous environments, often to reduce the risks to human workers. In the immediate future, it is likely that human workers will continue to use and direct these...

<https://arxiv.org/abs/2606.07211>

2 CONF

MoDA: Modulation Adapter for Fine-Grained Visual Grounding in Instructional MLLMs

ArXiv CS.AI 50%

arXiv:2506.01850v2 Announce Type: replace-cross Abstract: Multimodal Large Language Models (MLLMs) have achieved remarkable success in instruction-following tasks by integrating pretrained visual encoders with large language models (LLMs). However, existing approaches often...

<https://arxiv.org/abs/2506.01850>

3 CONF

AI Sovereignty: A Qualitative Model of Strategic Competition as AI Becomes an Instrument of National Power

ArXiv CS.AI 50%

arXiv:2606.07245v1 Announce Type: cross Abstract: AI sovereignty is the extent to which a nation independently controls its artificial intelligence (AI) technologies. The race toward ever-more-sophisticated frontier AI models is of increasing strategic importance, with nations...

<https://arxiv.org/abs/2606.07245>

4 CONF

The Geometry of Representational Failures in Vision Language Models

ArXiv CS.AI 50%

arXiv:2602.07025v2 Announce Type: replace-cross Abstract: Vision-Language Models (VLMs) exhibit puzzling failures in multi-object visual tasks, such as hallucinating non-existent elements or failing to identify the most similar objects among distractions. While these errors...

<https://arxiv.org/abs/2602.07025>

5 CONF

SleepExplain: Explainable Non-Rapid Eye Movement and Rapid Eye Movement Sleep Stage Classification from EEG Signal

ArXiv CS.AI 50%

arXiv:2606.07351v1 Announce Type: cross Abstract: Classification of sleep stages is one of the most important diagnostic approaches for a variety of sleep-related disorders. Electroencephalography (EEG) is regarded as a powerful tool for examining the association between...

<https://arxiv.org/abs/2606.07351>

6 CONF

A robust PPG foundation model using multimodal physiological supervision

ArXiv CS.AI 50%

arXiv:2606.07365v1 Announce Type: cross Abstract: Photoplethysmography (PPG), a non-invasive measure of changes in blood volume, is widely used in both wearable devices and clinical settings. Recent PPG foundation models either use open-source ICU datasets with pretraining...

<https://arxiv.org/abs/2606.07365>

7 CONF

The Masked Advantage: Uncovering Local-Language Access to Cultural Knowledge in LLMs

ArXiv CS.AI 50%

arXiv:2606.07422v1 Announce Type: cross Abstract: Large language models are increasingly used to answer culturally grounded questions across languages, yet it remains unclear whether local cultural knowledge is better accessed through English or the local language. Existing...

<https://arxiv.org/abs/2606.07422>

8 CONF

Watch, Remember, Reason: Human-View Video Understanding with MLLMs

ArXiv CS.AI 50%

arXiv:2606.07433v1 Announce Type: cross Abstract: Video understanding is being rapidly transformed by multimodal large language models (MLLMs), as research moves from short clips to long, multimodal, and knowledge-intensive video scenarios. These scenarios require models to...

<https://arxiv.org/abs/2606.07433>

9 CONF

TEVI: Text-Conditioned Editing of Visual Representations via Sparse Autoencoders for Improved Vision-Language Alignment

ArXiv CS.AI 50%

arXiv:2606.07451v1 Announce Type: cross Abstract: Vision-language models such as CLIP are highly useful for diverse tasks due to their shared image-text embedding space. Despite this, the image and text embeddings are often poorly aligned, affecting downstream performance....

<https://arxiv.org/abs/2606.07451>

0 CONF

PaperFlow: Profiling, Recommending, and Adapting Across Daily Paper Streams

ArXiv CS.AI 50%

arXiv:2606.07454v1 Announce Type: cross Abstract: Scientific paper recommendation is typically evaluated as static ranking over a fixed candidate set, yet real scientific reading unfolds as a daily, longitudinal process in which interests shift and feedback accumulates. We...

<https://arxiv.org/abs/2606.07454>

1 CONF

Graph Neural Network leveraging Higher-order Class Label Connectivity for Heterophilous Graphs

ArXiv CS.AI 50%

arXiv:2606.07475v1 Announce Type: cross Abstract: Node classification in graph neural networks (GNNs) has been widely applied in various fields of graph analysis. GNNs achieve high-accuracy node classification in homophilous graphs, where nodes with the same class label tend to...

<https://arxiv.org/abs/2606.07475>

2 CONF

Twelve quick tips for designing AI-driven HPC workflows

ArXiv CS.AI 50%

arXiv:2606.07491v1 Announce Type: cross Abstract: High-performance computing (HPC) clusters remain the backbone of large-scale scientific computation, traditionally executing deterministic, linear pipelines optimised for predictable performance. However, the pervasive...

<https://arxiv.org/abs/2606.07491>

3 CONF

How reliable are LLMs when it comes to playing dice?

ArXiv CS.AI 50%

arXiv:2606.07515v1 Announce Type: cross Abstract: We investigate the probabilistic reasoning capabilities of large language models through a controlled benchmarking study on discrete probability problems. We constructed two datasets, respectively a set of standard exercises and...

<https://arxiv.org/abs/2606.07515>

4 CONF

LLM-Augmented Digital Twin for Policy Evaluation in Short-Video Platforms

ArXiv CS.AI 50%

arXiv:2603.11333v2 Announce Type: replace Abstract: Short-video platforms are closed-loop, human-in-the-loop ecosystems where platform policy, creator incentives, and user behavior co-evolve. This feedback structure makes counterfactual policy evaluation difficult in production,...

<https://arxiv.org/abs/2603.11333>

5 CONF

EVA: Evolving Semantic Adversaries for Red-Teaming GUI Agents Against Environmental Injection Attacks

ArXiv CS.AI 50%

arXiv:2505.14289v2 Announce Type: replace Abstract: Graphical User Interface (GUI) agents powered by Multimodal Large Language Models (MLLMs) are increasingly deployed yet vulnerable to Environmental Injection Attacks (EIAs). However, current red-teaming methods are hindered by...

<https://arxiv.org/abs/2505.14289>

6 CONF

Exploring Flow-Lenia Universes with a Curiosity-driven AI Scientist: Discovering Diverse Ecosystem Dynamics

ArXiv CS.AI 50%

arXiv:2505.15998v4 Announce Type: replace Abstract: We present a curiosity-driven AI scientist method for discovering system-level dynamics in Flow-Lenia, a continuous cellular automaton (CA) with mass conservation and parameter localization. Building on prior work that uses...

<https://arxiv.org/abs/2505.15998>

7 CONF

Small Language Model Agents Enable Efficient and High-Quality Knowledge Mining

ArXiv CS.AI 50%

arXiv:2510.01427v3 Announce Type: replace Abstract: At the core of Deep Research is knowledge mining, the task of extracting structured information from massive unstructured text in response to user instructions. Large language models (LLMs) excel at interpreting such...

<https://arxiv.org/abs/2510.01427>

8 CONF

MHA-RAG: Improving Efficiency, Accuracy, and Consistency by Encoding Exemplars as Soft Prompts

ArXiv CS.AI 50%

arXiv:2510.05363v2 Announce Type: replace Abstract: Adapting Foundation Models to new domains with limited training data is challenging and computationally expensive. While prior work has demonstrated the effectiveness of using domain-specific exemplars as in-context...

<https://arxiv.org/abs/2510.05363>

9 CONF

TSAQA: Time Series Analysis Question And Answering Benchmark

ArXiv CS.AI 50%

arXiv:2601.23204v2 Announce Type: replace Abstract: Time series data are integral to critical applications across domains such as finance, healthcare, transportation, and environmental science. While recent work has begun to explore multi-task time series question answering...

<https://arxiv.org/abs/2601.23204>

00 CONF

Endogenous Resistance to Activation Steering in Language Models

ArXiv CS.AI 50%

arXiv:2602.06941v2 Announce Type: replace-cross Abstract: Large language models can recover mid-generation from task-misaligned activation steering, producing explicit verbal restarts (e.g., ``wait, that's not right'') and continuing on-topic even while the steering perturbation...

<https://arxiv.org/abs/2602.06941>

01 CONF

ReclAlm: A Multi-Agent Framework for Monitoring and Correcting Performance Decline in Medical Imaging AI

ArXiv CS.AI 50%

arXiv:2510.17004v2 Announce Type: replace-cross Abstract: Purpose: To develop and evaluate a multi-agent framework (ReclAlm) for automated monitoring, detection, and correction of performance decline in medical image classification models. Materials and Methods: ReclAlm is a...

<https://arxiv.org/abs/2510.17004>

02 CONF

Robust Instruction Compliance in Cooperative Multi-Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.12655v2 Announce Type: replace Abstract: Multi-agent reinforcement learning (MARL) in real-world use cases may need to adapt to external natural language instructions that interrupt ongoing behavior and conflict with long-horizon objectives. However, conditioning...

<https://arxiv.org/abs/2605.12655>

03 CONF

Beyond Output Matching: Preserving Internal Geometry in NVFP4 LLM Distillation

ArXiv CS.AI 50%

arXiv:2606.05682v2 Announce Type: replace Abstract: Demand for low-precision inference, including NVFP4-based approaches, has grown as large language models are increasingly deployed in latency and cost constrained production environments. Quantization-aware distillation (QAD)...

<https://arxiv.org/abs/2606.05682>

04 CONF

Should You Use Your Large Language Model to Explore or Exploit?

ArXiv CS.AI 50%

arXiv:2502.00225v4 Announce Type: replace-cross Abstract: We evaluate the ability of the current generation of large language models (LLMs) to help a decision-making agent facing an exploration-exploitation tradeoff. While previous work has largely study the ability of LLMs to...

<https://arxiv.org/abs/2502.00225>

05 CONF

Towards Efficient and Exact Forgetting Services in Pre-Trained-Model-based Continual Learning

ArXiv CS.AI 50%

arXiv:2505.12239v2 Announce Type: replace-cross Abstract: In Continual Learning (CL), using a Pre-Trained Model (PTM) as the feature extractor has become a popular practice. Accompanied by analytic classifiers, the PTM-based methods have achieved state-of-the-art performance in...

<https://arxiv.org/abs/2505.12239>

06 CONF

Bounded-Abstention Pairwise Learning to Rank

ArXiv CS.AI 50%

arXiv:2505.23437v2 Announce Type: replace-cross Abstract: Ranking systems influence decision-making in high-stakes domains like health, education, and employment, where they can have substantial economic and social impacts. This makes the integration of safety mechanisms...

<https://arxiv.org/abs/2505.23437>

07 CONF

CoQuIR: A Comprehensive Benchmark for Code Quality-Aware Information Retrieval

ArXiv CS.AI 50%

arXiv:2506.11066v3 Announce Type: replace-cross Abstract: Code retrieval is essential in modern software development, as it boosts code reuse and accelerates debugging. However, current benchmarks primarily emphasize functional relevance while neglecting critical dimensions of...

<https://arxiv.org/abs/2506.11066>

08 CONF

Telling stories, making Hanzi: AI-assisted co-creation with elderly migrants in urban China

ArXiv CS.AI 50%

arXiv:2507.01548v3 Announce Type: replace-cross Abstract: This paper explores how older migrants in urban China can record stories that everyday language and design often miss. We ran two co-creation workshops with 10 elders. Activities combined oral storytelling,...

<https://arxiv.org/abs/2507.01548>

09 CONF

Database Normalization via Dual-LLM Self-Refinement

ArXiv CS.AI 50%

arXiv:2508.17693v2 Announce Type: replace-cross Abstract: Database normalization is crucial to preserving data integrity. However, it is time-consuming and error-prone, as it is typically performed manually by data engineers. To this end, we present Miffie, a database...

<https://arxiv.org/abs/2508.17693>

10 CONF

Standard vs. Modular Sampling: Best Practices for Reliable LLM Unlearning

ArXiv CS.AI 50%

arXiv:2509.05316v2 Announce Type: replace-cross Abstract: A conventional LLM Unlearning setting consists of two subsets -"forget" and "retain", with the objectives of removing the undesired knowledge from the forget set while preserving the remaining knowledge from the retain....

<https://arxiv.org/abs/2509.05316>

11 CONF

MVCL-DAF++: Enhancing Multimodal Intent Recognition via Prototype-Aware Contrastive Alignment and Coarse-to-Fine Dynamic Attention Fusion

ArXiv CS.AI 50%

arXiv:2509.17446v3 Announce Type: replace-cross Abstract: Multimodal intent recognition (MMIR) suffers from weak semantic grounding and poor robustness under noisy or rare-class conditions. We propose MVCL-DAF++, which extends MVCL-DAF with two key modules: (1) Prototype-aware...

<https://arxiv.org/abs/2509.17446>

12 CONF

Proxy Reconstruction Pre-training for Ramp Flow Prediction at Highway Interchanges

ArXiv CS.AI 50%

arXiv:2510.03381v3 Announce Type: replace-cross Abstract: Interchanges are crucial nodes for vehicle transfers between highways, yet the lack of real-time ramp detectors creates blind spots in traffic prediction. To address this, we propose a Spatio-Temporal Decoupled...

<https://arxiv.org/abs/2510.03381>

13 CONF

MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models

ArXiv CS.AI 50%

arXiv:2510.11014v2 Announce Type: replace-cross Abstract: Autonomous robots often view rooms only partially, through a doorway, where the walls and scene structure hide the geometry and task-relevant semantics needed for safe navigation and goal-directed action. We ask whether...

<https://arxiv.org/abs/2510.11014>

14 CONF

It's a TRAP! Task-Redirecting Agent Persuasion Benchmark for Web Agents

ArXiv CS.AI 50%

arXiv:2512.23128v2 Announce Type: replace-cross Abstract: Web-based agents powered by large language models are increasingly used for tasks such as email management or professional networking. Their reliance on dynamic web content, however, makes them vulnerable to prompt...

<https://arxiv.org/abs/2512.23128>

15 CONF

Analysing Differences in Persuasive Language in LLM-Generated Text: Uncovering Stereotypical Gender Patterns

ArXiv CS.AI 50%

arXiv:2601.05751v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used for everyday communication tasks, including drafting interpersonal messages intended to influence and persuade. Prior work has shown that LLMs can successfully persuade...

<https://arxiv.org/abs/2601.05751>

16 CONF

Just-In-Time Reinforcement Learning: Continual Learning in LLM Agents Without Gradient Updates

ArXiv CS.AI 50%

arXiv:2601.18510v2 Announce Type: replace-cross Abstract: While Large Language Model (LLM) agents excel at general tasks, they inherently struggle with continual adaptation due to the frozen weights after deployment. Conventional reinforcement learning (RL) offers a solution but...

<https://arxiv.org/abs/2601.18510>

17 CONF

Enhancing Video Representations with Spatiotemporal-Semantic Residual to Mitigate Hallucinations in Video Large Multimodal Models

ArXiv CS.AI 50%

arXiv:2601.22574v2 Announce Type: replace-cross Abstract: Although Video Large Multimodal Models have achieved strong performance in video understanding, they still suffer from hallucination. Existing inference-time intervention methods usually modify videos under the...

<https://arxiv.org/abs/2601.22574>

18 CONF

Learning to Execute Graph Algorithms Exactly with Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2601.23207v2 Announce Type: replace-cross Abstract: Understanding what graph neural networks can learn, especially their ability to learn to execute algorithms, remains a central theoretical challenge. In this work, we prove exact learnability results for graph algorithms...

<https://arxiv.org/abs/2601.23207>

19 CONF

TokaMind: A Multi-Modal Transformer Foundation Model for Tokamak Plasma Dynamics

ArXiv CS.AI 50%

arXiv:2602.15084v2 Announce Type: replace-cross Abstract: We present TokaMind, to our knowledge the first open-source foundation model for tokamak plasma dynamics, based on a Multi-Modal Transformer (MMT) and pretrained on heterogeneous diagnostics from the publicly available...

<https://arxiv.org/abs/2602.15084>

20 CONF

EvoClaw: Evaluating AI Agents on Continuous Software Evolution

ArXiv CS.AI 50%

arXiv:2603.13428v2 Announce Type: replace-cross Abstract: With AI agents increasingly deployed as long-running systems, it becomes essential to autonomously construct and continuously evolve customized software to enable interaction within dynamic environments. Yet, existing...

<https://arxiv.org/abs/2603.13428>

21 CONF

Evaluating AI-based Scientific Knowledge Synthesis with Epidemiological Systematic Reviews

ArXiv CS.AI 50%

arXiv:2603.22327v2 Announce Type: replace-cross Abstract: Systematic literature reviews (SLRs) are a demanding and high-stakes form of scientific knowledge synthesis that remains underspecified as an evaluation setting for large language models (LLMs). We introduce AgentSLR, a...

<https://arxiv.org/abs/2603.22327>

22 CONF

More Capable, Less Cooperative? When LLMs Fail At Zero-Cost Collaboration

ArXiv CS.AI 50%

arXiv:2604.07821v2 Announce Type: replace-cross Abstract: Large language model (LLM) agents increasingly coordinate in multi-agent systems, yet we lack an understanding of where and why cooperation fails. Many real-world coordination problems are not social dilemmas: helping...

<https://arxiv.org/abs/2604.07821>

23 CONF

MACS: Modality-Aware Capacity Scaling for Efficient Multimodal MoE Inference

ArXiv CS.AI 50%

arXiv:2605.05225v3 Announce Type: replace-cross Abstract: Mixture-of-Experts Multimodal Large Language Models (MoE MLLMs) suffer from a significant efficiency bottleneck during Expert Parallelism (EP) inference due to the straggler effect. This issue is worsened in the...

<https://arxiv.org/abs/2605.05225>

24 CONF

COF26: A new on-top functional for multiconfiguration pair-density functional theory

ArXiv CS.AI 50%

arXiv:2605.06215v2 Announce Type: replace-cross Abstract: Multiconfiguration pair-density functional theory (MC-PDFT) provides an efficient and accurate framework for computing electronic energies in strongly correlated molecular systems, with the quality of the on-top...

<https://arxiv.org/abs/2605.06215>

25 CONF

Do Language Models Need Sleep? Offline Recurrence for Improved Online Inference

ArXiv CS.AI 50%

arXiv:2605.26099v3 Announce Type: replace-cross Abstract: Transformer-based large language models are increasingly used for long-horizon tasks; however, their attention mechanism scales poorly with context length. To handle this, we study a sleep-like consolidation mechanism in...

<https://arxiv.org/abs/2605.26099>

26 CONF

Scenario Generation for Risk-Aware Reinforcement Learning with Probably Approximately Safe Guarantees

ArXiv CS.AI 50%

arXiv:2606.04812v2 Announce Type: replace-cross Abstract: Guaranteeing safety is critical to the deployment of reinforcement learning (RL) agents in the real-world, especially as policies learned using deep RL may demonstrate susceptibility to transition perturbations that...

<https://arxiv.org/abs/2606.04812>